

**BABU BANARASI DAS
UNIVERSITY
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SCHOOL OF COMPUTER APPLICATIONS



**NoSQL Project
Employee Management System Using MongoDB**

Submitted to:

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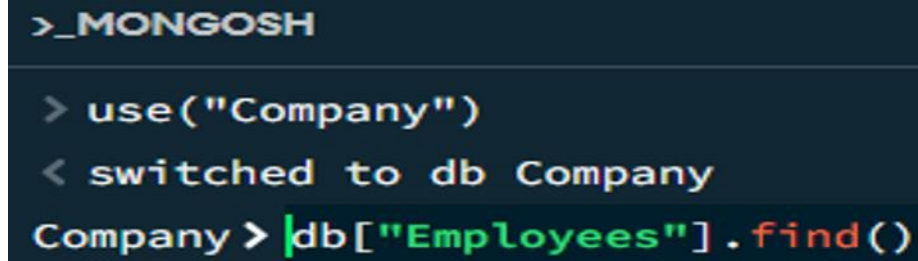
1. CREATE DATABASE

MongoDB Query:

use Company

Explanation: Creates or switches to the Company database.

Output Screenshot:



```
>_MONGOSH
> use("Company")
< switched to db Company
Company> db["Employees"].find()
```

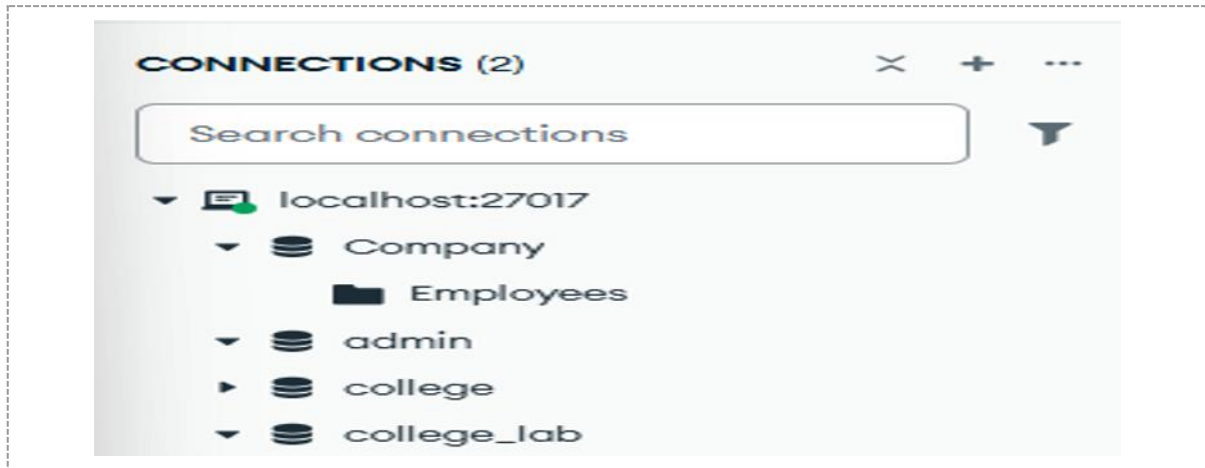
2. CREATE COLLECTION

MongoDB Query:

db.createCollection("employees")

Explanation: Creates a collection named employees.

Output Screenshot:



3. INSERT 50 EMPLOYEES

MongoDB Query:

```
db.employees.insertMany([
  {empId:1, name:"Aarav", age:23, salary:62000, department:"IT"},
  {empId:2, name:"Riya", age:26, salary:44628, department:"HR"},
  {empId:3, name:"Vihaan", age:24, salary:36717, department:"Operations"},
  {empId:4, name:"Sneha", age:30, salary:68698, department:"IT"},
  {empId:5, name:"Kabir", age:28, salary:78000, department:"IT"},
  {empId:6, name:"Diya", age:23, salary:45247, department:"HR"},
  {empId:7, name:"Reyansh", age:30, salary:31739, department:"Sales"},
  {empId:8, name:"Isha", age:30, salary:76925, department:"HR"},
  {empId:9, name:"Arjun", age:30, salary:44446, department:"Marketing"},
  {empId:10, name:"Meera", age:29, salary:48231, department:"HR"},
  {empId:11, name:"Aditya", age:22, salary:75753, department:"HR"},
  {empId:12, name:"Kavya", age:28, salary:48210, department:"Finance"},
  {empId:13, name:"Rohan", age:24, salary:92731, department:"HR"},
  {empId:14, name:"Tanya", age:27, salary:36078, department:"IT"},
  {empId:15, name:"Yash", age:28, salary:53526, department:"IT"},
  {empId:16, name:"Pooja", age:27, salary:47335, department:"Sales"},
  {empId:17, name:"Karan", age:22, salary:60108, department:"Operations"},
  {empId:18, name:"Neha", age:30, salary:93774, department:"IT"},
  {empId:19, name:"Siddharth", age:28, salary:66178, department:"IT"},
  {empId:20, name:"Divya", age:26, salary:91000, department:"Operations"},
  {empId:21, name:"Manav", age:27, salary:42601, department:"Sales"},
  {empId:22, name:"Simran", age:23, salary:73336, department:"IT"},
  {empId:23, name:"Rahul", age:25, salary:94567, department:"Finance"},
  {empId:24, name:"Ananya", age:23, salary:86785, department:"HR"},
  {empId:25, name:"Varun", age:26, salary:88000, department:"Marketing"},
  {empId:26, name:"Shreya", age:29, salary:84664, department:"Operations"},
  {empId:27, name:"Nikhil", age:27, salary:54260, department:"HR"},
  {empId:28, name:"Priya", age:27, salary:73920, department:"HR"},
  {empId:29, name:"Aryan", age:26, salary:91384, department:"Operations"},
  {empId:30, name:"Pallavi", age:23, salary:71613, department:"IT"},
  {empId:31, name:"Dev", age:24, salary:77784, department:"HR"},
  {empId:32, name:"Ritika", age:25, salary:60294, department:"HR"},
  {empId:33, name:"Sahil", age:28, salary:94849, department:"Finance"},
  {empId:34, name:"Anjali", age:30, salary:74866, department:"HR"},
  {empId:35, name:"Vivek", age:27, salary:45010, department:"IT"},
  {empId:36, name:"Kritika", age:22, salary:56290, department:"Finance"},
  {empId:37, name:"Harsh", age:26, salary:43826, department:"IT"},
  {empId:38, name:"Swati", age:27, salary:72954, department:"HR"},
  {empId:39, name:"Gaurav", age:29, salary:87978, department:"Marketing"},
  {empId:40, name:"Nidhi", age:29, salary:47359, department:"HR"},
  {empId:41, name:"Ishaan", age:24, salary:78823, department:"HR"},
  {empId:42, name:"Bhavna", age:30, salary:47219, department:"Sales"},
  {empId:43, name:"Suraj", age:28, salary:56175, department:"Sales"},
  {empId:44, name:"Payal", age:27, salary:39065, department:"HR"},
  {empId:45, name:"Akash", age:30, salary:35957, department:"Marketing"},
  {empId:46, name:"Sonal", age:22, salary:40016, department:"IT"},
```

```
{empId:47, name:"Raghav", age:24, salary:57666, department:"Operations"},  
{empId:48, name:"Tanvi", age:23, salary:55009, department:"Marketing"},  
{empId:49, name:"Devansh", age:29, salary:46476, department:"Sales"},  
{empId:50, name:"Kirti", age:30, salary:74583, department:"IT"}  
])
```

Explanation: Inserts multiple employee documents into the collection.

Output Screenshot:

```
acknowledged: true,
insertedIds: {
  '0': ObjectId('6ab8a71c1fd9b2ce912a38e0'),
  '1': ObjectId('6ab8a71c1fd9b2ce912a38e1'),
  '2': ObjectId('6ab8a71c1fd9b2ce912a38e2'),
  '3': ObjectId('6ab8a71c1fd9b2ce912a38e3'),
  '4': ObjectId('6ab8a71c1fd9b2ce912a38e4'),
  '5': ObjectId('6ab8a71c1fd9b2ce912a38e5'),
  '6': ObjectId('6ab8a71c1fd9b2ce912a38e6'),
  '7': ObjectId('6ab8a71c1fd9b2ce912a38e7'),
  '8': ObjectId('6ab8a71c1fd9b2ce912a38e8'),
  '9': ObjectId('6ab8a71c1fd9b2ce912a38e9'),
  '10': ObjectId('6ab8a71c1fd9b2ce912a38ea'),
  '11': ObjectId('6ab8a71c1fd9b2ce912a38eb'),
  '12': ObjectId('6ab8a71c1fd9b2ce912a38ec'),
  '13': ObjectId('6ab8a71c1fd9b2ce912a38ed'),
  '14': ObjectId('6ab8a71c1fd9b2ce912a38ee'),
  '15': ObjectId('6ab8a71c1fd9b2ce912a38ef'),
  '16': ObjectId('6ab8a71c1fd9b2ce912a38f0'),
  '17': ObjectId('6ab8a71c1fd9b2ce912a38f1'),
  '18': ObjectId('6ab8a71c1fd9b2ce912a38f2'),
  '19': ObjectId('6ab8a71c1fd9b2ce912a38f3'),
  '20': ObjectId('6ab8a71c1fd9b2ce912a38f4'),
  '21': ObjectId('6ab8a71c1fd9b2ce912a38f5'),
  '22': ObjectId('6ab8a71c1fd9b2ce912a38f6'),
  '23': ObjectId('6ab8a71c1fd9b2ce912a38f7'),
  '24': ObjectId('6ab8a71c1fd9b2ce912a38f8'),
  '25': ObjectId('6ab8a71c1fd9b2ce912a38f9'),
  '26': ObjectId('6ab8a71c1fd9b2ce912a38fa'),
  '27': ObjectId('6ab8a71c1fd9b2ce912a38fb'),
  '28': ObjectId('6ab8a71c1fd9b2ce912a38fc'),
  '29': ObjectId('6ab8a71c1fd9b2ce912a38fd'),
  '30': ObjectId('6ab8a71c1fd9b2ce912a38fe'),
  '31': ObjectId('6ab8a71c1fd9b2ce912a38ff'),
  '32': ObjectId('6ab8a71c1fd9b2ce912a3900'),
  '33': ObjectId('6ab8a71c1fd9b2ce912a3901'),
  '34': ObjectId('6ab8a71c1fd9b2ce912a3902'),
  '35': ObjectId('6ab8a71c1fd9b2ce912a3903'),
  '36': ObjectId('6ab8a71c1fd9b2ce912a3904'),
  '37': ObjectId('6ab8a71c1fd9b2ce912a3905'),
  '38': ObjectId('6ab8a71c1fd9b2ce912a3906'),
  '39': ObjectId('6ab8a71c1fd9b2ce912a3907'),
  '40': ObjectId('6ab8a71c1fd9b2ce912a3908'),
  '41': ObjectId('6ab8a71c1fd9b2ce912a3909'),
  '42': ObjectId('6ab8a71c1fd9b2ce912a390a'),
  '43': ObjectId('6ab8a71c1fd9b2ce912a390b'),
  '44': ObjectId('6ab8a71c1fd9b2ce912a390c'),
  '45': ObjectId('6ab8a71c1fd9b2ce912a390d'),
  '46': ObjectId('6ab8a71c1fd9b2ce912a390e'),
  '47': ObjectId('6ab8a71c1fd9b2ce912a390f'),
  '48': ObjectId('6ab8a71c1fd9b2ce912a3910'),
  '49': ObjectId('6ab8a71c1fd9b2ce912a3911')
}
```

4. DISPLAY ALL EMPLOYEES

MongoDB Query:

db.employees.find()

Explanation: Retrieves all documents from the employees collection.

Output Screenshot:

```
db.employees.find()
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e0'),
  empId: 1,
  name: 'Aarav',
  age: 23,
  salary: 62000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e1'),
  empId: 2,
  name: 'Riya',
  age: 26,
  salary: 44628,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e2'),
  empId: 3,
  name: 'Vihaan',
  age: 24,
  salary: 36717,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e3'),
  empId: 4,
  name: 'Sneha',
  age: 30,
  salary: 68698,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e4'),
  empId: 5,
  name: 'Kabir',
  age: 28,
  salary: 78000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e5'),
  empId: 6,
  name: 'Diya',
  age: 23,
  salary: 45247,
  department: 'HR'
}
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e6'),
  empId: 7,
  name: 'Reyansh',
  age: 30,
  salary: 31739,
  department: 'Sales'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e7'),
  empId: 8,
  name: 'Isha',
  age: 30,
  salary: 76925,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e8'),
  empId: 9,
  name: 'Arjun',
  age: 30,
  salary: 44446,
  department: 'Marketing'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e9'),
  empId: 10,
  name: 'Meera',
  age: 29,
  salary: 48231,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ea'),
  empId: 11,
  name: 'Aditya',
  age: 22,
  salary: 75753,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38eb'),
  empId: 12,
  name: 'Kavya',
  age: 28,
  salary: 48210,
  department: 'Finance'
}
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ee'),
  empId: 15,
  name: 'Yash',
  age: 28,
  salary: 53526,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ef'),
  empId: 16,
  name: 'Pooja',
  age: 27,
  salary: 47335,
  department: 'Sales'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f0'),
  empId: 17,
  name: 'Karan',
  age: 22,
  salary: 60108,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f1'),
  empId: 18,
  name: 'Neha',
  age: 30,
  salary: 93774,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f2'),
  empId: 19,
  name: 'Siddharth',
  age: 28,
  salary: 66178,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f3'),
  empId: 20,
  name: 'Divya',
  age: 26,
  salary: 91000,
  department: 'Operations'
}
```

Type "it" for more

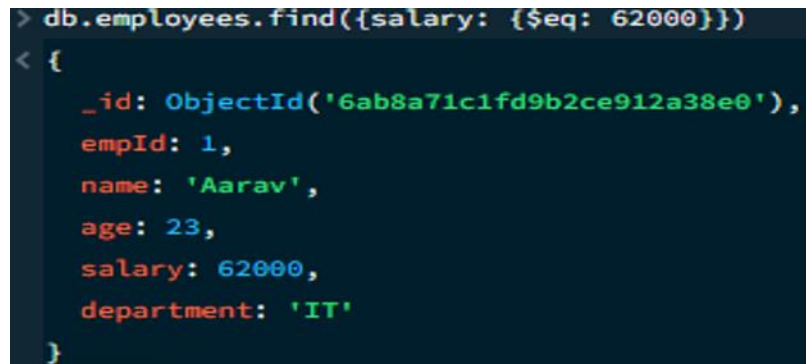
5. EQUAL TO — \$eq

MongoDB Query:

db.employees.find({salary: {\$eq: 62000}})

Explanation: Finds employees whose salary is exactly 62000.

Output Screenshot:



```
> db.employees.find({salary: {$eq: 62000}})
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e0'),
  empId: 1,
  name: 'Aarav',
  age: 23,
  salary: 62000,
  department: 'IT'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

db.employees.find({salary: {\$gt: 85000}})

Explanation: Finds employees whose salary is greater than 85000.

Output Screenshot:

```

db.employees.find({salary: {$gt: 85000}})
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ec'),
  empId: 13,
  name: 'Rohan',
  age: 24,
  salary: 92731,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f1'),
  empId: 18,
  name: 'Neha',
  age: 30,
  salary: 93774,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f3'),
  empId: 20,
  name: 'Divya',
  age: 26,
  salary: 91000,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f6'),
  empId: 23,
  name: 'Rahul',
  age: 25,
  salary: 94567,
  department: 'Finance'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f7'),
  empId: 24,
  name: 'Ananya',
  age: 23,
  salary: 86785,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f8'),
  empId: 25,
  name: 'Varun',
  age: 26,
  salary: 88000,
  department: 'Marketing'
}

```

```

{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fc'),
  empId: 29,
  name: 'Aryan',
  age: 26,
  salary: 91384,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3900'),
  empId: 33,
  name: 'Sahil',
  age: 28,
  salary: 94849,
  department: 'Finance'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3906'),
  empId: 39,
  name: 'Gaurav',
  age: 29,
  salary: 87978,
  department: 'Marketing'
}

```

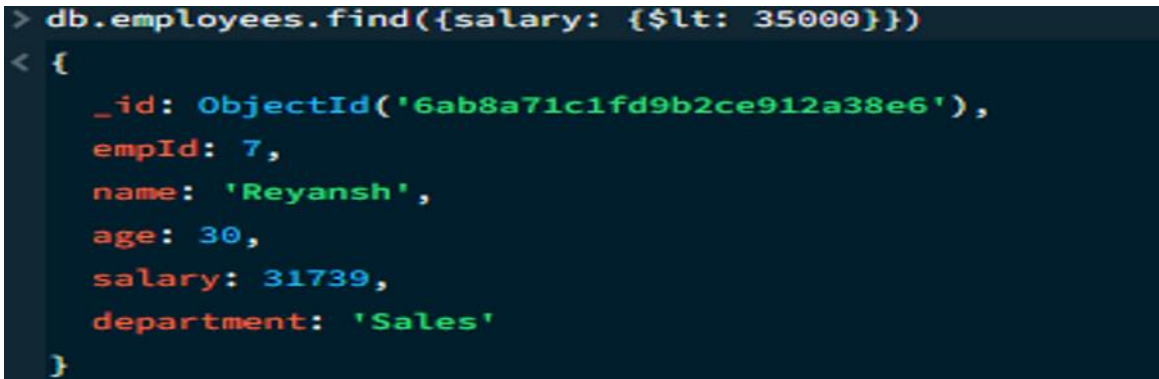
7. LESS THAN — \$lt

MongoDB Query:

db.employees.find({salary: {\$lt: 35000}})

Explanation: Finds employees whose salary is less than 35000.

Output Screenshot:



```
> db.employees.find({salary: {$lt: 35000}})
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e6'),
  empId: 7,
  name: 'Reyansh',
  age: 30,
  salary: 31739,
  department: 'Sales'
}
```

The screenshot shows a MongoDB shell session. The first line is the command `> db.employees.find({salary: {$lt: 35000}})`. The second line is the result, which is a JSON object: `< { _id: ObjectId('6ab8a71c1fd9b2ce912a38e6'), empId: 7, name: 'Reyansh', age: 30, salary: 31739, department: 'Sales' }`. The result is displayed in a dark-themed terminal window.

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

db.employees.find({salary: {\$gte: 90000}})

Explanation: Finds employees whose salary is 90000 or above.

Output Screenshot:

```
db.employees.find({salary: {$gte: 90000}})
[
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38ec'),
    empId: 13,
    name: 'Rohan',
    age: 24,
    salary: 92731,
    department: 'HR'
  },
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38f1'),
    empId: 18,
    name: 'Neha',
    age: 30,
    salary: 93774,
    department: 'IT'
  },
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38f3'),
    empId: 20,
    name: 'Divya',
    age: 26,
    salary: 91000,
    department: 'Operations'
  },
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38f6'),
    empId: 23,
    name: 'Rahul',
    age: 25,
    salary: 94567,
    department: 'Finance'
  },
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38fc'),
    empId: 29,
    name: 'Aryan',
    age: 26,
    salary: 91384,
    department: 'Operations'
  },
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a3900'),
    empId: 33,
    name: 'Sahil',
    age: 28,
    salary: 94849,
    department: 'Finance'
  }
]
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

`db.employees.find({salary: {$lte: 40000}})`

Explanation: Finds employees whose salary is 40000 or below.

Output Screenshot:

```
> db.employees.find({salary: {$lte: 40000}})
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e2'),
  empId: 3,
  name: 'Vihaan',
  age: 24,
  salary: 36717,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e6'),
  empId: 7,
  name: 'Reyansh',
  age: 30,
  salary: 31739,
  department: 'Sales'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ed'),
  empId: 14,
  name: 'Tanya',
  age: 27,
  salary: 36078,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a390b'),
  empId: 44,
  name: 'Payal',
  age: 27,
  salary: 39065,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a390c'),
  empId: 45,
  name: 'Akash',
  age: 30,
  salary: 35957,
  department: 'Marketing'
}
```

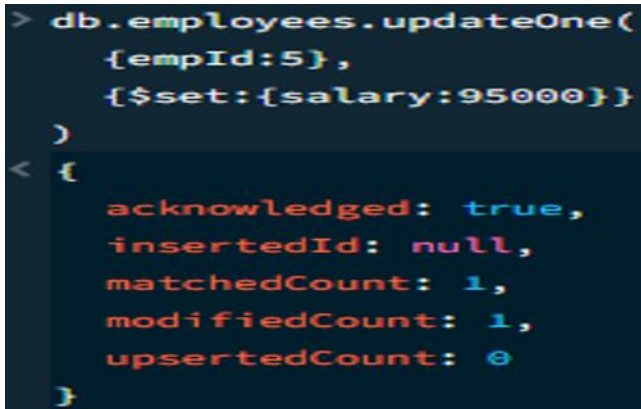
10. UPDATE OPERATION

MongoDB Query:

```
db.employees.updateOne(  
  {empId:5},  
  {$set:{salary:95000}}  
)
```

Explanation: Updates the salary to 95000 of the employee with empId 5.

Output Screenshot:



```
> db.employees.updateOne(  
  {empId:5},  
  {$set:{salary:95000}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

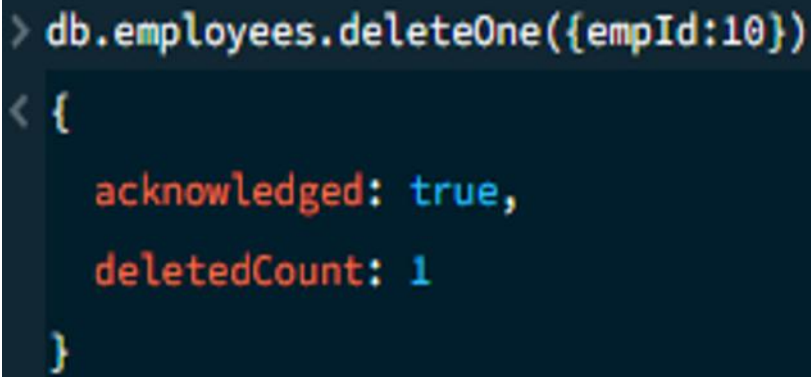
11. DELETE ONE OPERATION

MongoDB Query:

db.employees.deleteOne({empId:10})

Explanation: Deletes one document matching empId 10.

Output Screenshot:



```
> db.employees.deleteOne({empId:10})
< {
  acknowledged: true,
  deletedCount: 1
}
```

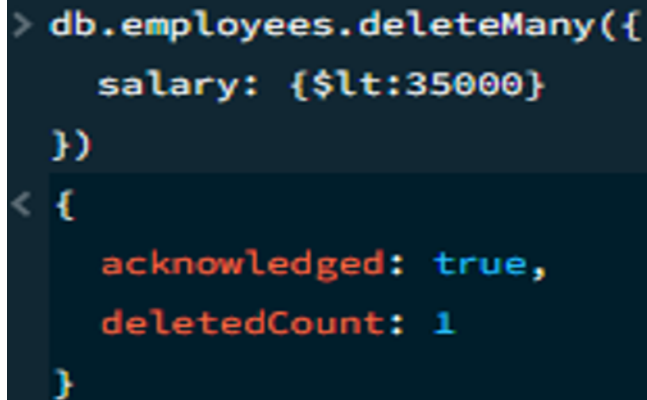
12. DELETE MANY OPERATION

MongoDB Query:

```
db.employees.deleteMany({  
  salary: {$lt:35000}  
})
```

Explanation: Deletes all employees having a salary below 35000.

Output Screenshot:



```
> db.employees.deleteMany({  
  salary: {$lt:35000}  
})  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

13. AND OPERATION — \$and

MongoDB Query:

```
db.employees.find({
  $and:[
    {salary:{$gt:80000}},
    {age:26}
  ]
})
```

Explanation: Finds employees satisfying both conditions.

Output Screenshot:



```
> db.employees.find({
  $and:[
    {salary:{$gt:80000}},
    {age:26}
  ]
})
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f3'),
  empId: 20,
  name: 'Divya',
  age: 26,
  salary: 91000,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f8'),
  empId: 25,
  name: 'Varun',
  age: 26,
  salary: 88000,
  department: 'Marketing'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fc'),
  empId: 29,
  name: 'Aryan',
  age: 26,
  salary: 91384,
  department: 'Operations'
}
```

14. OR OPERATION — \$or

MongoDB Query:

```
db.employees.find({  
  $or:[  
    {department:"IT"},  
    {department:"HR"}  
  ]  
})
```

Explanation: Finds employees who work in the IT or HR department.

Output Screenshot:

```

> db.employees.find({
  $or:[
    {department:"IT"},
    {department:"HR"}
  ]
})
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e0'),
  empId: 1,
  name: 'Aarav',
  age: 23,
  salary: 62000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e1'),
  empId: 2,
  name: 'Riya',
  age: 26,
  salary: 44628,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e3'),
  empId: 4,
  name: 'Sneha',
  age: 30,
  salary: 68698,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e4'),
  empId: 5,
  name: 'Kabir',
  age: 28,
  salary: 95000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e5'),
  empId: 6,
  name: 'Diya',
  age: 23,
  salary: 45247,
  department: 'HR'
}

```

```

{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e7'),
  empId: 8,
  name: 'Isha',
  age: 30,
  salary: 76925,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ea'),
  empId: 11,
  name: 'Aditya',
  age: 22,
  salary: 75753,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ec'),
  empId: 13,
  name: 'Rohan',
  age: 24,
  salary: 92731,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ed'),
  empId: 14,
  name: 'Tanya',
  age: 27,
  salary: 36078,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ee'),
  empId: 15,
  name: 'Vash',
  age: 28,
  salary: 53526,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f1'),
  empId: 18,
  name: 'Neha',
  age: 30,
  salary: 93774,
  department: 'IT'
}

```

```

{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f2'),
  empId: 19,
  name: 'Siddharth',
  age: 28,
  salary: 66178,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f5'),
  empId: 22,
  name: 'Simran',
  age: 23,
  salary: 73336,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f7'),
  empId: 24,
  name: 'Ananya',
  age: 23,
  salary: 86785,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fa'),
  empId: 27,
  name: 'Nikhil',
  age: 27,
  salary: 54260,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fb'),
  empId: 28,
  name: 'Priya',
  age: 27,
  salary: 73920,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fd'),
  empId: 30,
  name: 'Pallavi',
  age: 23,
  salary: 71613,
  department: 'IT'
}

```

```

{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fe'),
  empId: 31,
  name: 'Dev',
  age: 24,
  salary: 77784,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ff'),
  empId: 32,
  name: 'Ritika',
  age: 25,
  salary: 60294,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3901'),
  empId: 34,
  name: 'Anjali',
  age: 30,
  salary: 74866,
  department: 'HR'
}

```

Type "it" for more

15. SORT OPERATION

A. MongoDB Query:

db.employees.find().sort({salary: 1})

Explanation: Sorts employees by salary from lowest to highest. Use 1 for ascending order.

Output Screenshot:

```
db.employees.find().sort({salary: 1})
[
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a390c'),
    empId: 45,
    name: 'Akash',
    age: 30,
    salary: 35957,
    department: 'Marketing'
  }
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38ed'),
    empId: 14,
    name: 'Tanya',
    age: 27,
    salary: 36078,
    department: 'IT'
  }
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38e2'),
    empId: 3,
    name: 'Vihaan',
    age: 24,
    salary: 36717,
    department: 'Operations'
  }
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a390b'),
    empId: 44,
    name: 'Payal',
    age: 27,
    salary: 39065,
    department: 'HR'
  }
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a390d'),
    empId: 46,
    name: 'Sonal',
    age: 22,
    salary: 40016,
    department: 'IT'
  }
  {
    _id: ObjectId('6ab8a71c1fd9b2ce912a38f4'),
    empId: 21,
    name: 'Manav',
    age: 27,
    salary: 42601,
    department: 'Sales'
  }
]
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3904'),
  empId: 37,
  name: 'Harsh',
  age: 26,
  salary: 43826,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e8'),
  empId: 9,
  name: 'Arjun',
  age: 30,
  salary: 44446,
  department: 'Marketing'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e1'),
  empId: 2,
  name: 'Riya',
  age: 26,
  salary: 44628,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3902'),
  empId: 35,
  name: 'Vivek',
  age: 27,
  salary: 45010,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e5'),
  empId: 6,
  name: 'Diya',
  age: 23,
  salary: 45247,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3910'),
  empId: 49,
  name: 'Devansh',
  age: 29,
  salary: 46476,
  department: 'Sales'
}
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3909'),
  empId: 42,
  name: 'Bhavna',
  age: 30,
  salary: 47219,
  department: 'Sales'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ef'),
  empId: 16,
  name: 'Pooja',
  age: 27,
  salary: 47335,
  department: 'Sales'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3907'),
  empId: 40,
  name: 'Nidhi',
  age: 29,
  salary: 47359,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38eb'),
  empId: 12,
  name: 'Kavya',
  age: 28,
  salary: 48210,
  department: 'Finance'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ee'),
  empId: 15,
  name: 'Yash',
  age: 28,
  salary: 53526,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fa'),
  empId: 27,
  name: 'Nikhil',
  age: 27,
  salary: 54260,
  department: 'HR'
}
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a390f'),
  empId: 48,
  name: 'Tanvi',
  age: 23,
  salary: 55009,
  department: 'Marketing'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a390a'),
  empId: 43,
  name: 'Suraj',
  age: 28,
  salary: 56175,
  department: 'Sales'
}
Type "it" for more
```

B. MongoDB Query:

db.employees.find().sort({salary: -1})

Explanation: Sorts employees by salary from highest to lowest. Use -1 for descending order.

Output Screenshot:

```

> db.employees.find().sort({salary: -1})
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e4'),
  empId: 5,
  name: 'Kabir',
  age: 28,
  salary: 95000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3900'),
  empId: 33,
  name: 'Sahil',
  age: 28,
  salary: 94849,
  department: 'Finance'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f6'),
  empId: 23,
  name: 'Rahul',
  age: 25,
  salary: 94567,
  department: 'Finance'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f1'),
  empId: 18,
  name: 'Neha',
  age: 30,
  salary: 93774,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ec'),
  empId: 13,
  name: 'Rohan',
  age: 24,
  salary: 92731,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fc'),
  empId: 29,
  name: 'Aryan',
  age: 26,
  salary: 91384,
  department: 'Operations'
}

```

```

{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f3'),
  empId: 20,
  name: 'Divya',
  age: 26,
  salary: 91000,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f8'),
  empId: 25,
  name: 'Varun',
  age: 26,
  salary: 88000,
  department: 'Marketing'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3906'),
  empId: 39,
  name: 'Gaurav',
  age: 29,
  salary: 87978,
  department: 'Marketing'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f7'),
  empId: 24,
  name: 'Ananya',
  age: 23,
  salary: 86785,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f9'),
  empId: 26,
  name: 'Shreya',
  age: 29,
  salary: 84664,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3908'),
  empId: 41,
  name: 'Ishaan',
  age: 24,
  salary: 78823,
  department: 'HR'
}

```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e7'),
  empId: 8,
  name: 'Isha',
  age: 30,
  salary: 76925,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ea'),
  empId: 11,
  name: 'Aditya',
  age: 22,
  salary: 75753,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3901'),
  empId: 34,
  name: 'Anjali',
  age: 30,
  salary: 74866,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3911'),
  empId: 50,
  name: 'Kirti',
  age: 30,
  salary: 74583,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fb'),
  empId: 28,
  name: 'Priya',
  age: 27,
  salary: 73920,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f5'),
  empId: 22,
  name: 'Simran',
  age: 23,
  salary: 73336,
  department: 'IT'
}
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3905'),
  empId: 38,
  name: 'Swati',
  age: 27,
  salary: 72954,
  department: 'HR'
}
Type "it" for more
```

16. LIMIT OPERATION

MongoDB Query:

db.employees.find().limit(5)

Explanation: Displays only the first five documents.

Output Screenshot:

```
> db.employees.find().limit(5)
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e0'),
  empId: 1,
  name: 'Aarav',
  age: 23,
  salary: 62000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e1'),
  empId: 2,
  name: 'Riya',
  age: 26,
  salary: 44628,
  department: 'HR'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e2'),
  empId: 3,
  name: 'Vihaan',
  age: 24,
  salary: 36717,
  department: 'Operations'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e3'),
  empId: 4,
  name: 'Sneha',
  age: 30,
  salary: 68698,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e4'),
  empId: 5,
  name: 'Kabir',
  age: 28,
  salary: 95000,
  department: 'IT'
}
```

17. SEARCH BY DEPARTMENT

MongoDB Query:

db.employees.find({department:"IT"})

Explanation: Finds all employees who work in the IT department.

Output Screenshot:

```
> db.employees.find({department:"IT"})
```

```
< {
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e0'),
  empId: 1,
  name: 'Aarav',
  age: 23,
  salary: 62000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e3'),
  empId: 4,
  name: 'Sneha',
  age: 30,
  salary: 68698,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38e4'),
  empId: 5,
  name: 'Kabir',
  age: 28,
  salary: 95000,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ed'),
  empId: 14,
  name: 'Tanya',
  age: 27,
  salary: 36078,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38ee'),
  empId: 15,
  name: 'Yash',
  age: 28,
  salary: 53526,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f1'),
  empId: 18,
  name: 'Neha',
  age: 30,
  salary: 93774,
  department: 'IT'
}
```

```
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f2'),
  empId: 19,
  name: 'Siddharth',
  age: 28,
  salary: 66178,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38f5'),
  empId: 22,
  name: 'Simran',
  age: 23,
  salary: 73336,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a38fd'),
  empId: 30,
  name: 'Pallavi',
  age: 23,
  salary: 71613,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3902'),
  empId: 35,
  name: 'Vivek',
  age: 27,
  salary: 45010,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a3904'),
  empId: 37,
  name: 'Harsh',
  age: 26,
  salary: 43826,
  department: 'IT'
}
{
  _id: ObjectId('6ab8a71c1fd9b2ce912a390d'),
  empId: 46,
  name: 'Sonal',
  age: 22,
  salary: 40016,
  department: 'IT'
}
```

```
{  
  _id: ObjectId('6ab8a71c1fd9b2ce912a3911'),  
  empId: 50,  
  name: 'Kirti',  
  age: 30,  
  salary: 74583,  
  department: 'IT'  
}
```

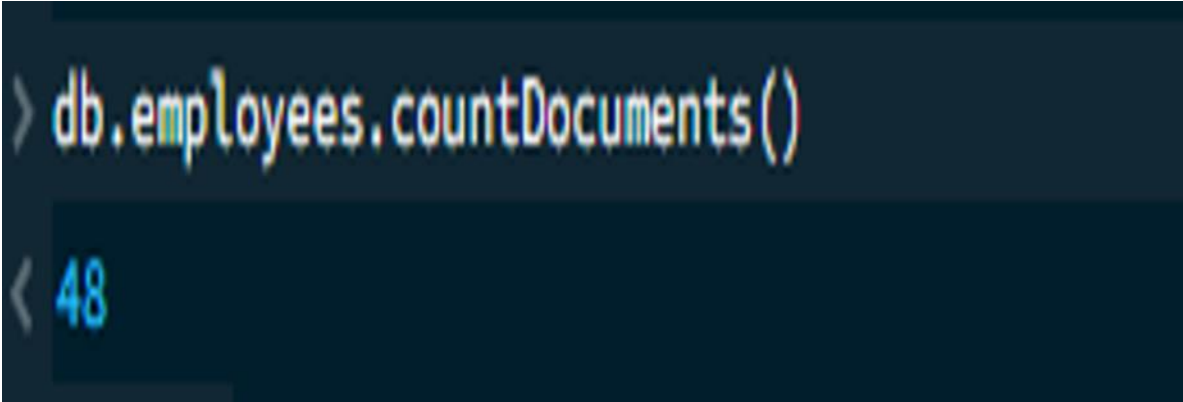
18. COUNT DOCUMENTS

MongoDB Query:

db.employees.countDocuments()

Explanation: Counts documents currently present in the collection (performed after the delete operations).

Output Screenshot:



```
> db.employees.countDocuments()  
  
< 48
```

MONGODB OPERATORS USED

Operator	Meaning	Example
\$eq	Equal to	salary: {\$eq:62000}
\$gt	Greater than	salary: {\$gt:85000}
\$lt	Less than	salary: {\$lt:35000}
\$gte	Greater than or equal to	salary: {\$gte:90000}
\$lte	Less than or equal to	salary: {\$lte:40000}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. An Employee Management System containing employee information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to arrange data in ascending and descending order, how comparison operators such as \$eq, \$gt, \$lt, \$gte, and \$lte work, along with logical operators such as \$and and \$or.

Overall, this project helped me understand how MongoDB can be used to manage and query employee data efficiently.